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DESIGN AND ANALYSIS OF MEMS ACCELEROMETER FOR SPACE LAUNCH VEHICLE APPLICATION

A PROJECT REPORT

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(Autonomous Institution affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report “**DESIGN AND ANALYSIS OF MEMS ACCELEROMETER FOR SPACE LAUNCH VEHICLE APPLICATION**” is the bonafide work of “**MOHAMED SALMAN AL FARISH A (927623BEC135), MOHAMMED AATHIL J (927623BEC136), MUKESH M (927623BEC141)** who carried out the project work during the academic year 2025- 2026 under my supervision.

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INTERNAL EXAMINER

EXTERNAL EXAMINER

VISION AND MISSION OF THE INSTITUTION

VISION

To emerge as a leader among the top institutions in the field of technical education.

MISSION

- Produce smart technocrats with empirical knowledge who can surmount the global challenges.
- Create a diverse, fully-engaged learner-centric campus environment to provide quality education to the students.
- Maintain mutually beneficial partnerships with our alumni, industry and professional associations.

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

VISION

To empower the Electronics and Communication Engineering students with emerging technologies, professionalism, innovative research and social responsibility.

MISSION

- M1: Attain the academic excellence through innovative teaching learning process, research areas & laboratories and Consultancy projects.
- M2: Inculcate the students in problem solving and lifelong learning ability.
- M3: Provide entrepreneurial skills and leadership qualities.
- M4: Render the technical knowledge and skills of faculty members.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1: Core Competence: Graduates will have a successful career in academia or industry associated with Electronics and Communication Engineering.

PEO2: Professionalism: Graduates will provide feasible solutions for the challenging problems through comprehensive research and innovation in the allied areas of Electronics and Communication Engineering.

PEO3: Lifelong Learning: Graduates will contribute to the social needs through lifelong learning, practicing professional ethics and leadership quality.

PROGRAM OUTCOMES(POs)

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: Applying knowledge in various areas, like Electronics, Communications, Signal processing, VLSI, Embedded systems etc., in the design and implementation of Engineering application.

PSO2: Able to solve complex problems in Electronics and Communication Engineering with analytical and managerial skills either independently or in team using latest hardware and software tools to fulfil the industrial expectations.

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ABSTRACT

Micro-Electro-Mechanical Systems (MEMS) accelerometers play a crucial role in modern aerospace applications due to their compact size, high sensitivity, and low power consumption. This project focuses on the design and analysis of a MEMS-based accelerometer intended for space launch vehicle applications. The proposed structure consists of a centrally suspended proof mass supported by symmetric micro-scale suspension beams, which deform when external acceleration is applied. The displacement of the proof mass is directly proportional to the applied acceleration, enabling accurate motion sensing. The accelerometer model is designed and analyzed using COMSOL Multiphysics to study displacement behavior, stress distribution, and structural stability. Advanced materials such as Gallium Nitride (GaN) are considered to enhance mechanical strength and reliability under extreme launch conditions. The simulation results demonstrate that the proposed MEMS accelerometer provides high sensitivity and stable performance, making it suitable for applications such as launch vehicle navigation, vibration monitoring, and aerospace inertial sensing systems.

Abstract (Key words)	POs Mapping
MEMS Accelerometer, Proof Mass Structure, Gallium Nitride (GaN), COMSOL Multiphysics, Space Launch Vehicle, Inertial Sensing.	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2

SDG Goal		Remarks
SDG 9 	This project supports sustainable technological development by promoting advanced MEMS sensing technology for aerospace systems. The proposed accelerometer improves reliability and precision in space launch vehicle monitoring, contributing to innovative infrastructure and high-performance engineering solutions.	Supports technological innovation in aerospace sensing systems by developing a compact and high-precision MEMS accelerometer. The project contributes to improved monitoring and reliability of space launch vehicles through advanced micro-sensor technology.

Project Component	Relevant IEEE Standards
MEMS Accelerometer Sensor	IEEE 2700
Inertial Measurement Systems	IEEE 1559
Aerospace Electronic Systems	IEEE 1220
Structural/Vibration Monitoring	IEEE 1451
Simulation & Software Validation	IEEE 29119

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LIST OF ABBREVIATIONS

ACRONYM	ABBREVIATION
MEMS	Micro-Electro-Mechanical Systems
COMSOL	COMSOL Multiphysics Simulation Software
Si	Silicon
GaN	Gallium Nitride
SiC	Silicon Carbide
GaAs	Gallium Arsenide

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF THE PROJECT

Micro-Electro-Mechanical Systems (MEMS) technology has become an important part of modern sensing and control systems due to its compact size, low power consumption, and high sensitivity. MEMS accelerometers are widely used to measure acceleration, vibration, and motion in various fields such as aerospace, automotive systems, robotics, and consumer electronics. In aerospace applications, especially in space launch vehicles, accurate acceleration measurement is critical for guidance, navigation, and structural monitoring. During launch operations, the vehicle experiences extremely high vibration, shock loads, and rapid acceleration that must be monitored precisely to ensure mission safety and stability. MEMS accelerometers provide a suitable solution for these conditions because they are small, lightweight, and capable of delivering accurate measurements under dynamic environments. By integrating advanced materials and optimized structural design, MEMS accelerometers can offer improved mechanical strength and sensing accuracy for high-performance aerospace applications.

1.2 MOTIVATION FOR THE STUDY

The development of advanced sensing technologies is essential for improving the reliability and safety of aerospace systems. Conventional accelerometers used in large-scale systems often suffer from disadvantages such as bulky size, higher power consumption, and limited compatibility with modern miniaturized electronics. These limitations make them less suitable for compact aerospace systems where weight, power efficiency, and precision are critical factors. MEMS-based accelerometers provide a promising alternative because they can be fabricated in miniature sizes while maintaining high sensitivity and

fast response. The motivation for this study is to design and analyze a MEMS accelerometer capable of accurately measuring acceleration under extreme conditions such as those experienced in space launch vehicles. By utilizing simulation tools such as COMSOL Multiphysics, the project aims to understand the structural behavior, displacement response, and stress distribution of the accelerometer structure. This study contributes toward developing reliable, compact, and high-performance sensing devices suitable for aerospace monitoring systems.

1.2.1 PROBLEM STATEMENT

Space launch vehicles experience extreme mechanical conditions including high vibration, shock loads, and rapid acceleration during launch and stage separation. Accurate measurement of these forces is essential for ensuring proper guidance, navigation, and structural integrity of the vehicle. However, conventional accelerometers often face limitations such as larger size, high power requirements, and reduced sensitivity when exposed to harsh environments. These challenges create the need for a compact, highly sensitive, and mechanically stable accelerometer that can operate reliably under aerospace conditions. Therefore, this project focuses on designing a MEMS-based accelerometer with a centrally suspended proof mass structure capable of accurately detecting acceleration through displacement measurement. The proposed design is analyzed using COMSOL Multiphysics to evaluate its performance in terms of displacement, stress distribution, and mechanical stability under applied acceleration forces.

1.2.2 OBJECTIVES OF THE PROJECT

The objective of this project is to design and analyze a MEMS-based accelerometer capable of accurately measuring acceleration in aerospace applications. The proposed device utilizes a centrally suspended proof mass supported by micro-scale suspension beams that deform under applied acceleration. Another objective is to achieve high sensitivity and mechanical stability while maintaining a compact and low-power structure suitable for space launch vehicle environments. The project also aims to study the displacement behavior, stress distribution, and structural performance of the accelerometer using COMSOL Multiphysics simulation. By optimizing the structural design and material selection, the project seeks to demonstrate a reliable and efficient MEMS accelerometer that can be used for guidance, navigation, and vibration monitoring in aerospace systems.

1.2.3 SCOPE OF THE PROJECT

The scope of this project includes the design, modeling, and simulation of a MEMS accelerometer structure using COMSOL Multiphysics. The system focuses on analyzing the mechanical behavior of the accelerometer when external acceleration forces are applied. The project primarily investigates parameters such as proof mass displacement, stress distribution, and structural stability of the suspension beams. The design is intended for aerospace applications, particularly for monitoring acceleration in space launch vehicles. The scope is limited to simulation-based analysis rather than physical fabrication of the device. However, the proposed design framework can be extended in the future to include fabrication, experimental testing, and integration with electronic sensing circuits for real-world applications.

1.2.4 METHODOLOGY OVERVIEW

The methodology of this project begins with designing the MEMS accelerometer structure consisting of a central proof mass, suspension beams, and fixed anchor points. The geometry of the accelerometer is created and defined within the COMSOL Multiphysics environment. Appropriate material properties such as density, Young's modulus and Poisson's ratio are assigned to the structure. Solid Mechanics physics is then applied to simulate the mechanical response of the device under external acceleration. Time-dependent or static analysis is performed to observe the displacement of the proof mass and the stress distribution along the suspension beams. The simulation results are analyzed to evaluate the sensitivity and mechanical stability of the accelerometer design. Through this systematic approach, the project demonstrates the feasibility of using MEMS accelerometers for precise acceleration measurement in aerospace environments.

CHAPTER 2

LITERATURE REVIEW

2.1 LITERATURE OVERVIEW

In recent years, Micro-Electro-Mechanical Systems (MEMS) technology has gained significant attention in the field of sensing and measurement systems. MEMS accelerometers have become an important component in many applications such as automotive safety systems, consumer electronics, industrial monitoring and aerospace navigation. Their compact size, low power consumption and high sensitivity make them suitable for applications where traditional macro-scale sensors are impractical.

Researchers have focused on improving the structural design, sensitivity and mechanical stability of MEMS accelerometers. Various structural configurations such as cantilever beams, folded springs and centrally suspended proof masses have been explored to enhance performance. Modern simulation tools such as COMSOL Multiphysics allow accurate modelling of these microstructures before fabrication. Through simulation, parameters such as displacement, stress distribution and resonance frequency can be analyzed to optimize the accelerometer design.

In aerospace systems, particularly in Space Launch Vehicle, accelerometers play a crucial role in guidance, navigation and control systems. Accurate measurement of acceleration ensures that the launch vehicle follows the correct trajectory during lift-off and stage separation. Therefore, the development of robust MEMS accelerometers capable of operating under extreme vibration and shock conditions has become an important research area.

2.2 RELATED WORK

2.2.1 MEMS ACCELEROMETER STRUCTURE DESIGN

Several studies have investigated different structural designs to improve the performance of MEMS accelerometers. Early MEMS accelerometers were mainly based on simple cantilever beam structures where a proof mass was attached to a flexible beam. When acceleration is applied, the proof mass moves, creating measurable displacement or capacitance change.

Later research introduced folded beam suspension systems that provide better mechanical stability and improved sensitivity. The centrally suspended proof mass design distributes stress evenly across the suspension beams, reducing the risk of structural failure during high vibration conditions. These structural improvements help the accelerometer maintain accuracy even under harsh environments such as aerospace launch conditions.

2.2.2 SIMULATION AND MODELLING USING COMSOL

Simulation plays a critical role in the design and analysis of MEMS devices. Researchers widely use **COMSOL Multiphysics** to model MEMS accelerometer structures and analyze their mechanical behavior. The software enables the integration of multiple physics modules such as solid mechanics, electrostatics and structural mechanics.

In many studies, the accelerometer geometry is first created in the simulation environment and material properties such as density, Young's modulus and Poisson's ratio are assigned. External acceleration forces are then applied to observe proof mass displacement and stress distribution in the suspension beams.

These simulations help designers optimize the geometry and improve the sensitivity and reliability of the accelerometer before fabrication.

2.2.3 MEMS ACCELEROMETERS IN AEROSPACE APPLICATIONS

MEMS accelerometers are widely used in aerospace applications due to their small size and high precision. In launch vehicles and satellites, accelerometers are integrated into the inertial navigation system to measure linear acceleration and detect motion changes.

During the launch phase of a Space Launch Vehicle, the vehicle experiences extreme acceleration and vibration. Accurate sensing of these forces is essential for maintaining stability and ensuring correct trajectory control. Researchers continue to explore new materials, improved suspension structures and optimized proof mass designs to enhance the performance of MEMS accelerometers in such demanding environments.

CHAPTER 3

EXISTING SYSTEM

Conventional accelerometers have been widely used for measuring acceleration in various applications such as automotive safety systems, industrial monitoring, and aerospace navigation. Traditional accelerometers are usually macro-scale devices made using mechanical components such as springs, masses, and piezoelectric sensors. These devices convert mechanical motion into electrical signals to measure acceleration. Although they provide accurate measurements, they are generally bulky, consume more power, and are difficult to integrate into compact electronic systems.

In aerospace applications such as space launch vehicles, sensors must withstand extreme conditions including high vibration, shock, and rapid acceleration changes. Traditional accelerometers often face challenges in maintaining accuracy under such harsh environments. As a result, researchers have focused on developing MEMS-based accelerometers that offer improved performance with smaller size and lower power consumption.

3.1 Limitations of Conventional Accelerometers

Traditional accelerometers are relatively large in size due to their mechanical construction, making them unsuitable for compact aerospace systems where space and weight are critical factors. These sensors also consume higher electrical power compared to modern MEMS-based sensors. Additionally, their mechanical components may experience wear and performance degradation when exposed to continuous vibration or shock conditions during launch operations.

Another limitation is their reduced sensitivity when measuring very small acceleration changes. Conventional sensors may struggle to detect fine variations

in motion, which are essential for precise navigation and control systems in aerospace applications.

3.2 Challenges in Existing Accelerometer Designs

Existing accelerometer designs often face several challenges related to mechanical stability and environmental conditions. During the launch of a space vehicle, extremely high acceleration forces and vibrations occur. These conditions can cause structural deformation or signal instability in traditional accelerometers. As a result, the measurement accuracy may decrease, affecting the performance of navigation systems.

Another challenge is integrating the sensor with modern electronic systems. Traditional accelerometers require additional signal conditioning circuits and bulky mounting structures, which increase the complexity and cost of the system.

3.2.1 Lack of Compact and Lightweight Design

Many conventional accelerometers are not designed with miniaturization in mind. Their larger size and weight make them difficult to integrate into small electronic modules used in modern aerospace systems. This limitation restricts their use in applications where compact and lightweight components are required.

3.2.2 Limited Sensitivity and Mechanical Reliability

Traditional accelerometers often provide limited sensitivity when measuring small acceleration changes. Additionally, their mechanical structures may not withstand extreme vibration or shock conditions experienced during rocket launch. These limitations highlight the need for improved sensor designs that offer higher sensitivity, better mechanical stability, and reliable performance under harsh operating environments.

CHAPTER 4

PROBLEM SYSTEM

In modern aerospace systems such as space launch vehicles, accurate acceleration measurement is essential for guidance, navigation, and control operations. During launch, the vehicle experiences extreme vibration, high shock loads, and rapid acceleration changes. Conventional accelerometers used in earlier systems are often bulky, consume more power, and lack the sensitivity required to measure small variations in acceleration. These limitations reduce the efficiency and reliability of measurement systems in harsh aerospace environments.

Additionally, traditional sensors may suffer from mechanical wear, reduced stability, and difficulty in integration with compact electronic systems. Their large size and higher power consumption make them unsuitable for modern aerospace technologies where lightweight and energy-efficient components are required.

To address these challenges, a compact and highly sensitive sensing device is required. MEMS-based accelerometers provide a promising solution due to their small size, low power consumption, and ability to detect very small acceleration changes with high precision. By using micro-scale structures such as proof masses and suspension beams, MEMS accelerometers can provide reliable acceleration measurement even under extreme conditions experienced during space launch operations.

CHAPTER 5

PROPOSED SYSTEM

The proposed system presents the design and simulation of a Micro-Electro-Mechanical Systems (MEMS) based accelerometer intended for aerospace applications such as space launch vehicles. The aim of the system is to measure acceleration accurately under extreme environmental conditions including vibration, shock, and high dynamic forces experienced during launch operations. Conventional accelerometers are often bulky and consume higher power, making them unsuitable for compact aerospace electronic systems.

The proposed MEMS accelerometer consists of a central proof mass suspended by micro-scale beams connected to fixed anchor points on a substrate. When external acceleration is applied to the device, the inertial force causes the proof mass to move slightly from its equilibrium position. This displacement is proportional to the applied acceleration. The displacement can then be measured and converted into an electrical signal for accurate acceleration sensing.

The structure of the accelerometer is designed and simulated using COMSOL Multiphysics, which enables detailed modelling of the mechanical behaviour of the microstructure. Material properties such as density, Young's modulus, and Poisson's ratio are assigned to the structure. Solid Mechanics physics is applied to analyze the displacement of the proof mass and the stress distribution across the suspension beams when acceleration is applied.

The proposed design offers several advantages including compact size, low power consumption, high sensitivity, and improved mechanical stability. These features make the MEMS accelerometer suitable for use in Space Launch Vehicle systems where precise motion sensing is essential.

5.1 System Architecture Overview

The system architecture of the MEMS accelerometer consists of a substrate, suspension beams, proof mass, and anchor structures. The proof mass is placed at the center of the structure and is connected to the substrate through flexible suspension beams. These beams allow controlled movement of the proof mass when acceleration is applied.

When the system experiences acceleration, the proof mass undergoes displacement due to inertial forces. The suspension beams deform slightly to allow this movement while maintaining structural stability. The displacement generated by the proof mass is proportional to the applied acceleration force. This mechanical displacement can then be detected through sensing mechanisms such as capacitive sensing or piezoresistive sensing.

The entire structure is modeled in COMSOL Multiphysics, where simulations are performed to analyze displacement, stress distribution, and mechanical reliability. The results help evaluate the sensitivity and stability of the accelerometer design before fabrication.

5.2 ANALYTICAL DESIGN AND MATERIAL WISE CALCULATION

The proposed MEMS accelerometer works on the basis of deformation of suspension beams with the applied external force on the central proof mass. The analytical behavior of the structure can be explained using beam theory.

A. General Analytical Formulation

Let,

- l = beam length
- b = beam width
- t = beam thickness
- E = Young's modulus of the material
- I = moment of inertia of the beam
- F = applied force
- δ = beam deflection

The moment of inertia of the beam is given by,

$$I = \frac{bt^3}{12}$$

The beam stiffness is expressed as,

$$k = \frac{Ebt^3}{L^3}$$

The deflection of the beam is,

$$\delta = \frac{FL^3}{12EI}$$

Since the beam geometry and applied load remain the same for all materials,

$$\delta \propto \frac{1}{E}$$

This shows that the displacement is inversely proportional to Young's modulus.

B. Silicon Material

For Silicon,

- Density $\rho_{Si} = 2329 \text{ kg/m}^3$
- Young's modulus $E_{Si} = 170 \times 10^9 \text{ Pa}$
- Poisson's ratio $\nu_{Si} = 0.28$

The beam deflection is

$$\delta_{Si} = \frac{FL^3}{12E_{Si}I}$$

Substituting the Young's modulus,

$$\delta_{Si} = \frac{FL^3}{12(170 \times 10^9)I}$$

Silicon will provide moderate levels of displacement while striking a good balance between sensitivity and mechanical strength.

C. Gallium Arsenide Material

For Gallium Arsenide,

- Density $\rho_{GaAs} = 5316 \text{ kg/m}^3$
- Young's modulus $E_{GaAs} = 85.98 \times 10^9 \text{ Pa}$
- Poisson's ratio $\nu_{GaAs} = 0.31$

The beam deflection is

$$\delta_{GaAs} = \frac{FL^3}{12E_{GaAs}I}$$

Substituting the Young's modulus,

$$\delta_{GaAs} = \frac{FL^3}{12(85.98 \times 10^9)I}$$

Gallium arsenide is more susceptible to displacement than silicon due to its lower Young's modulus; hence, it also has higher sensitivity.

D. Gallium Nitride Material

For Gallium Nitride,

- Density $\rho_{GaN} = 6070 \text{ kg/m}^3$
- Young's modulus $E_{GaN} = 300 \times 10^9 \text{ Pa}$
- Poisson's ratio $\nu_{GaN} = 0.21$

The beam deflection is

$$\delta_{GaN} = \frac{FL^3}{12E_{GaN}I}$$

Substituting the Young's modulus,

$$\delta_{GaN} = \frac{FL^3}{12(300 \times 10^9)I}$$

Since it is much stiffer than other materials, Gallium Nitride has lower displacements and is more structurally stable than any other material.

E. Silicon Carbide Material

For Silicon Carbide,

- Density $\rho_{SiC} = 3216 \text{ kg/m}^3$
- Young's modulus $E_{SiC} = 470 \times 10^9 \text{ Pa}$
- Poisson's ratio $\nu_{SiC} = 0.17$

The beam deflection is

$$\delta_{SiC} = \frac{FL^3}{12E_{SiC}I}$$

Substituting the Young's modulus,

$$\delta_{SiC} = \frac{FL^3}{12(470 \times 10^9)I}$$

Due to its extremely high stiffness, Silicon Carbide possesses the lowest amount of deformation out of all of the materials being compared.

F. Displacement Trend Based on Material Stiffness

Since

$$\delta \propto \frac{1}{E}$$

the displacement order is

$$\delta_{GaAs} > \delta_{Si} > \delta_{GaN} > \delta_{SiC}$$

This analytical result supports the simulation results obtained from COMSOL.

G. Material Efficiency Interpretation

The performance and suitability of accelerometer materials used in MEMS applications are subject to availability of such materials for a certain application.

1. GaAs will give the maximum amount of displacement capability combined with the best sensitivity for any application.
2. Silicon will provide moderate amounts of displacement combined with a good trade-off between performance and reliability.
3. GaN will give the maximum amount of stiffness with a long lifespan of reliability; and
4. SiC will provide the maximum rigidity for use under harsh environments.

Higher strength materials like GaN and SiC are the best choice for use as the main structural material for space rockets; while GaAs is best for high sensitivity accelerometers.

5.3 Block Diagram

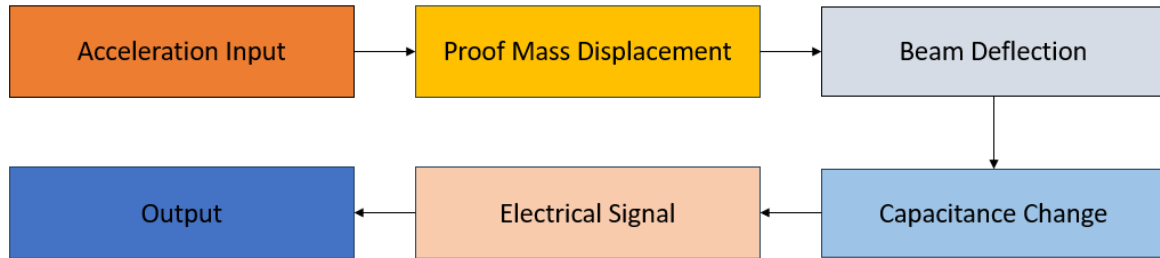


Fig 5.1: Block Diagram

Description:

The block diagram illustrates the working principle of the proposed MEMS accelerometer system. When external acceleration is applied to the device, it acts as the input to the system. This acceleration causes the central proof mass to move slightly from its original position due to inertial forces. As the proof mass displaces, the suspension beams connected to it experience mechanical deflection. This beam deflection results in a change in the capacitance between the sensing electrodes of the accelerometer structure. The variation in capacitance is then converted into an electrical signal using appropriate sensing circuitry. Finally, the electrical signal is processed to produce the output, which represents the measured acceleration. This mechanism enables accurate detection of acceleration in applications such as space launch vehicle navigation and vibration monitoring.

CHAPTER 6

SIMULATION AND IMPLEMENTATION

The proposed MEMS accelerometer design is modeled and analyzed using COMSOL Multiphysics simulation software. The structure consists of a centrally suspended proof mass connected to four suspension beams that are anchored to fixed supports. This configuration allows the proof mass to move freely when external acceleration is applied. The simulation helps analyze the mechanical behavior of the accelerometer under acceleration conditions.

The Solid Mechanics module in COMSOL is used to simulate the deformation and stress distribution of the structure. Material properties such as density, Young's modulus, and Poisson's ratio are assigned to the model. Boundary conditions are defined by fixing the anchor points while allowing the proof mass to displace under the applied acceleration.

Different semiconductor materials such as Silicon (Si), Gallium Arsenide (GaAs), Gallium Nitride (GaN), and Silicon Carbide (SiC) are used in the simulation to evaluate their performance for MEMS accelerometer applications. hygienic and hazardous environments.

6.1 Displacement Analysis

Displacement analysis is carried out to study the movement of the proof mass when acceleration is applied to the accelerometer structure. The displacement magnitude indicates the sensitivity of the device to external forces. The simulation results show that the maximum displacement occurs at the central proof mass, while the displacement gradually decreases toward the suspension beams and fixed anchor points. This behavior confirms that the suspension beams effectively transfer the inertial force generated by acceleration to the sensing

structure. When different materials are analyzed, it is observed that Silicon exhibits moderate displacement with stable structural behavior. Gallium Arsenide produces slightly higher displacement due to its relatively lower stiffness compared to silicon. Gallium Nitride demonstrates controlled displacement with good mechanical stability, making it suitable for high-performance MEMS applications. Silicon Carbide shows the lowest displacement because of its high mechanical stiffness and structural strength. The displacement distribution clearly shows the bending of suspension beams while the proof mass experiences maximum movement under acceleration.

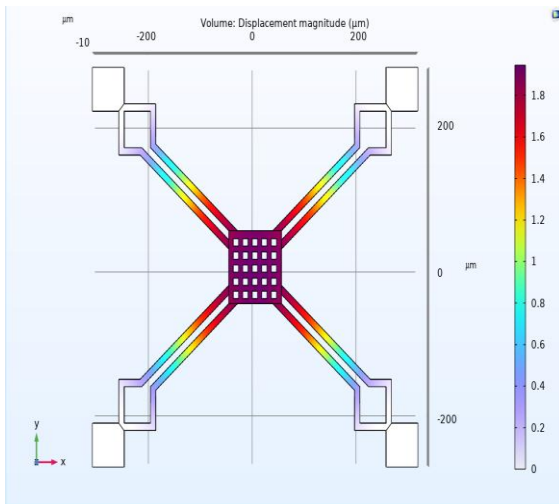


Fig 6.1.1: Silicon Material

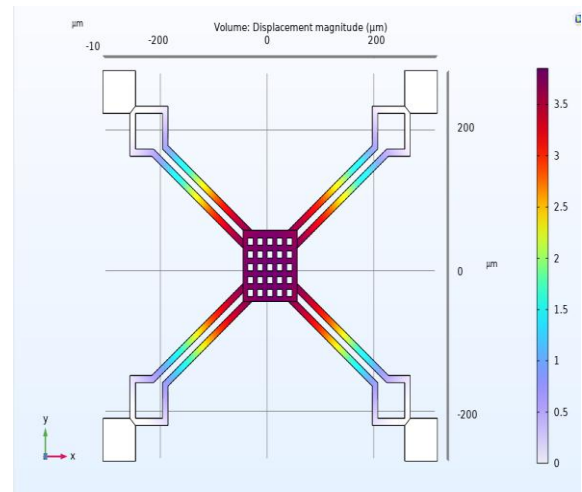


Fig 6.1.2 : Gallium arsenide

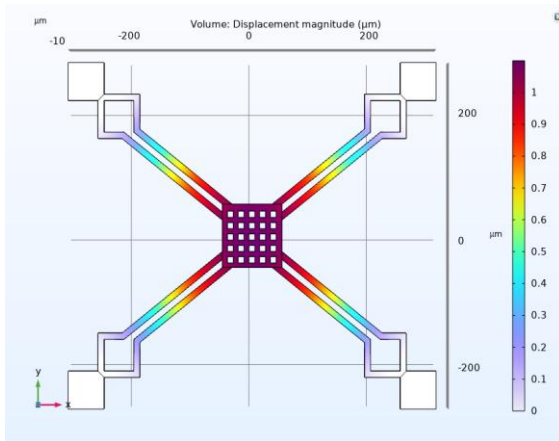
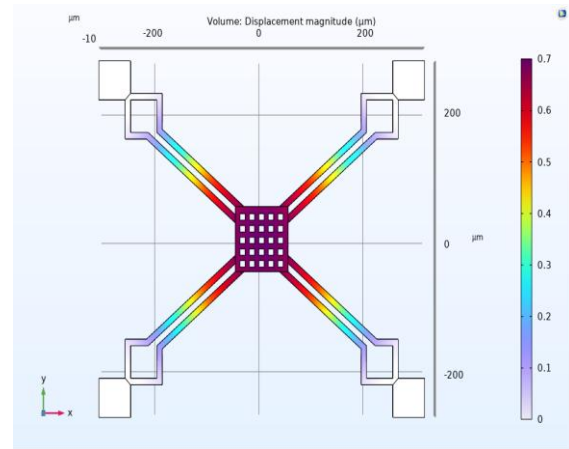


Fig 6.1.3 : Gallium Nitride (GaN)



**Fig 6.1.4: Silicon Carbide – SiC
(6H)**

6.2 Strain Analysis

Strain analysis is performed to examine the deformation of the suspension beams and structural components when the accelerometer experiences acceleration. This analysis helps identify the regions where mechanical stress is concentrated and ensures that the structure can withstand the applied forces without failure. The simulation results indicate that the maximum strain occurs near the junction between the proof mass and the suspension beams, as these regions experience the highest bending during displacement. The strain gradually reduces toward the anchor points where the beams are fixed. When comparing different materials, Silicon provides balanced strain distribution and reliable structural stability. Gallium Arsenide experiences slightly higher strain due to its lower mechanical strength. Gallium Nitride shows moderate strain values with good resistance to deformation. Silicon Carbide exhibits the lowest strain values because of its high rigidity and mechanical robustness. These results confirm that the proposed accelerometer design can safely operate under the applied acceleration conditions.

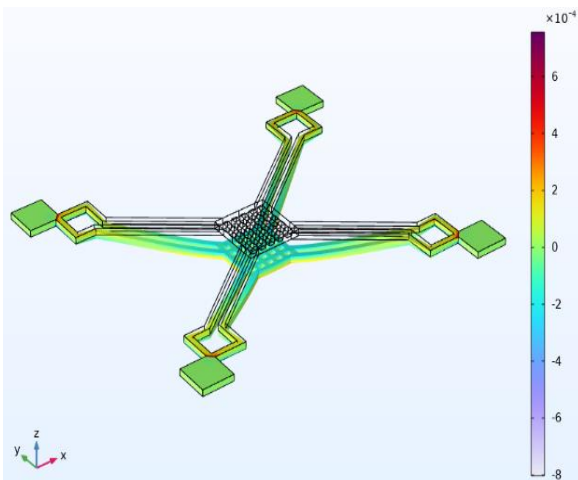


Fig 6.2.1: Silicon Material

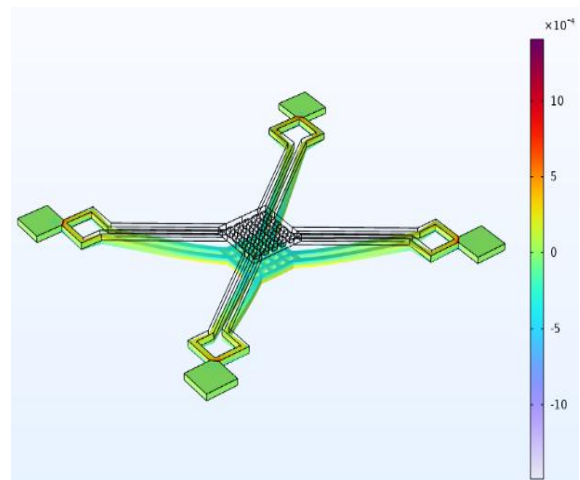


Fig 6.2.2: Gallium arsenide

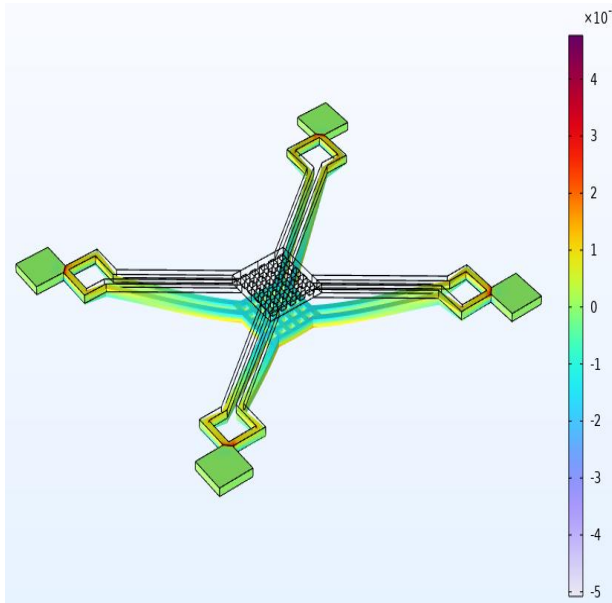


Fig 6.2.3: Gallium Nitride (GaN)

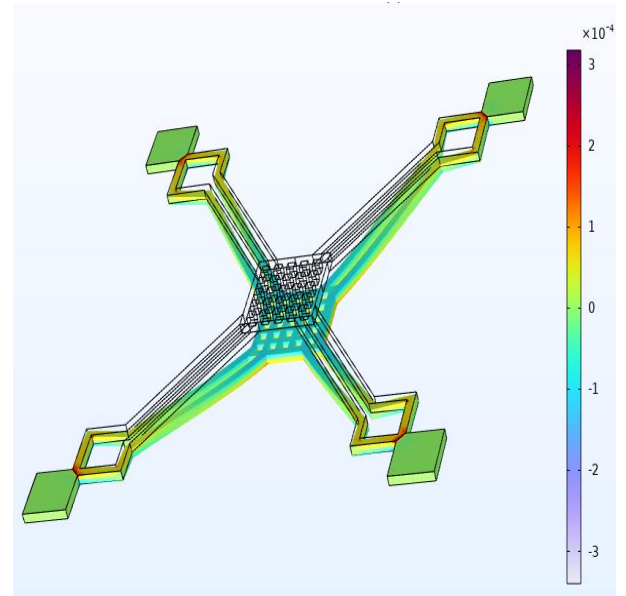


Fig 6.2.4: Silicon Carbide – (6H)

IMPLEMENTATION

The proposed MEMS accelerometer is implemented using COMSOL Multiphysics simulation software. The structure consists of a central proof mass connected to suspension beams and fixed anchor points. Different materials such as Silicon, Gallium Arsenide, Gallium Nitride, and Silicon Carbide are assigned to the model to analyze their performance. The Solid Mechanics module is used to simulate the response of the structure when acceleration is applied. Displacement and strain analyses are performed to observe proof mass movement and beam deformation. The results help evaluate the sensitivity and mechanical stability of the accelerometer design.

CHAPTER 7

RESULTS AND DISCUSSION

The proposed MEMS accelerometer structure was designed and analyzed using COMSOL Multiphysics simulation software. When external acceleration was applied to the structure, the simulation results showed that the central proof mass experienced noticeable displacement while the suspension beams underwent controlled deformation. The displacement was highest at the proof mass and gradually decreased toward the fixed anchor points. This behavior confirms that the suspension beams effectively transfer inertial forces and allow the proof mass to move proportionally to the applied acceleration, enabling accurate acceleration sensing.

Further analysis was carried out using different semiconductor materials such as Silicon (Si), Gallium Arsenide (GaAs), Gallium Nitride (GaN), and Silicon Carbide (SiC). The results indicated that Silicon provided balanced displacement and strain distribution suitable for MEMS fabrication, while Gallium Arsenide showed slightly higher displacement with increased strain. Gallium Nitride demonstrated good mechanical stability with moderate displacement, and Silicon Carbide exhibited the lowest displacement due to its high stiffness. These results confirm that the proposed accelerometer design is capable of providing reliable acceleration measurement and can be effectively used in aerospace applications such as space launch vehicle navigation and vibration monitoring.

CHAPTER 8

CONCLUSION AND FUTURE WORK

CONCLUSION

The proposed MEMS accelerometer was successfully designed and analyzed using COMSOL Multiphysics simulation software. The structure consisting of a central proof mass, suspension beams, and fixed anchor points was modeled to study its mechanical response under applied acceleration. Simulation results showed that the proof mass undergoes measurable displacement while the suspension beams experience controlled deformation, enabling accurate acceleration sensing. The displacement and strain analyses confirmed that the structure maintains mechanical stability while responding effectively to external forces.

Different semiconductor materials such as Silicon (Si), Gallium Arsenide (GaAs), Gallium Nitride (GaN), and Silicon Carbide (SiC) were also analyzed to evaluate their performance. The results demonstrated that each material exhibits different mechanical characteristics influencing sensitivity and structural strength. Among the tested materials, Silicon and Gallium Nitride showed balanced performance suitable for MEMS accelerometer applications. Overall, the proposed design demonstrates reliable performance and is suitable for applications such as space launch vehicle navigation, vibration monitoring, and inertial sensing systems.

FUTURE WORK

Future work can focus on improving the sensitivity and performance of the MEMS accelerometer by optimizing the geometry of the proof mass and suspension beams. Advanced sensing mechanisms such as capacitive or piezoresistive sensing circuits can be integrated to convert mechanical displacement into precise electrical signals. Fabrication of the MEMS accelerometer using microfabrication techniques can also be explored to validate the simulation results experimentally.

Further research may involve implementing multi-axis accelerometer designs capable of measuring acceleration in three dimensions. The use of advanced materials with higher thermal and mechanical stability can also enhance performance in harsh environments such as aerospace applications. With these improvements, the MEMS accelerometer can be further developed into a highly reliable sensing device suitable for space launch vehicles, satellite systems, and other high-precision navigation technologies.

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